

Sensor-Grounded Admission: Polity Receipts with Attested Substrate State

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Abstract

A federated admission kernel decides whether to admit a signed receipt into its polity’s history. The decision is well-defined when the kernel’s sensing posture is healthy; it is silent when the kernel has lost a sensor mid-decision. Existing constructions treat the substrate as honestly-sensing-by-assumption: receipts produced under degraded sensors are indistinguishable from receipts produced under healthy ones. We give an admission predicate that is conditioned on a signed attestation, embedded in every receipt, of which sensors were installed, active, healthy, and degraded at decision time, with dropped-event and deadline-miss counts. The polity’s admission relation evaluates this attestation against a constitution-declared required set; admission under a degraded substrate is structurally distinct from admission under a healthy substrate, and the structural distinction lives at the predicate level rather than at the heuristic alert level. The headline result is the existence of two receipts sharing identical body bytes whose distinct sensor attestations discharge the admission predicate to opposite verdicts. The supporting results connect the structural distinction to the parent admission kernel’s five-mode trust ladder: the partition-contingency mode is decidable as the case where attested-healthy providers are a strict subset of policy-required providers; destructive admission requires healthy attestation of every receipt-backed-floor sensor; an amendment that re-admits a receipt previously admitted under a degraded substrate must carry a re-admission witness. The retired assumption is the substrate-honest-by-assumption row: the kernel now carries a falsifiable signed statement about its own sensing posture, and the verifier’s admission predicate evaluates that statement rather than presuming it.

1 Introduction

A federated admission kernel decides whether to admit a signed receipt into its polity’s history under a constitution. The decision is sound when the kernel’s sensing posture is healthy: the receipt body reflects what the kernel observed, the kernel observed what its sensors reported, and the sensors were live, configured, and not dropping events. The decision is silent when any of those conditions fails. The kernel’s signature on the receipt body says nothing about whether the body reflects a complete observation or an observation made with the relevant sensor absent, degraded, or backlogged. A receipt produced under a partial substrate looks identical, on the wire, to a receipt produced under a complete one.

The parent admission construction in the Chio substrate makes this assumption explicit and names it as residual risk: the verifier-owned trust stores and the kernel are honest by assumption with respect to their own sensing posture [5]. The assumption is a row in the substrate’s assumption ledger, not a consequence of any of its theorems, and the gap is not addressed structurally. The threat model controls the agent, the foreign polity, and the malformed evidence. It does not control the kernel’s sensors. A kernel that loses one sensor mid-decision and produces a receipt with the

missing sensor’s contribution silently elided cannot be distinguished from one that kept the sensor live; the receiver admits both.

This paper closes the gap with a primitive that is not new in kind, only in placement: every receipt carries, alongside its body, a signed attestation of the kernel’s sensing posture at decision time. The attestation lists each registered sensor’s installed, active, healthy, and degraded flags, plus dropped-event and deadline-miss counts. The attestation is canonicalized into the same DSSE subject digest as the receipt body, so a verifier cannot accept the body without accepting the attestation. The admission predicate is then conditioned on the attestation: a constitution declares a required sensor set, and the admission relation succeeds only when the attestation shows the required set healthy.

The structural distinction is the contribution. Admission under a degraded substrate is not heuristically suspect; it is in a different ladder mode than admission under a healthy one, decidable on the attestation field. The partition-contingency mode in the parent paper’s five-mode ladder, where the parent paper carries a placeholder, becomes the structurally-defined mode reached exactly when the attested-healthy providers are a strict sublist of the policy-required providers. Destructive admission, where the parent paper conditions only on ladder rank as a heuristic floor, becomes type-conditioned on healthy attestation of every receipt-backed-floor sensor.

The retired assumption is the substrate-honest-by-assumption row in the parent paper. It is not retired by an additional cryptographic primitive but by a placement choice: the falsifiable statement now lives inside the receipt rather than outside it. A kernel that lies about its sensors is now signing a falsifiable claim, which is the precondition for any external sensor-coverage auditor to disagree. A kernel that does not lie continues to admit receipts under the same predicate, which is now decidable on a substrate-attested field. The strict strengthening is conditional on the existence of such an out-of-band sensor-coverage auditor; the auditor is the structural complement of this work and is named as future work in Section 10.

Threat model. The adversary controls the agent that produces the receipt body and can therefore submit arbitrary body content for admission. The adversary can observe but not modify the substrate’s attestation signing key, which is rooted in a trusted-execution environment whose root-of-trust assumption the construction inherits from the parent substrate [5]. Within those bounds the adversary may replay signed receipts and attestations across decision contexts, and may attempt to forge attestation contents that pass the constitution’s sensor-set check. The adversary cannot compromise the attestation signing key (the single-key-collapse residual is named in Section 9), cannot suppress sensor-level reporting from a sensor that is itself healthy, and cannot alter a signed attestation after signing without invalidating the signature. The verifier holds the substrate’s attestation public key through the same trust-store discipline that supplies receipt-body verification keys in the parent construction. The construction protects the verifier of a receipt rather than the operator of the signing kernel: the falsifiable statement that the construction adds is a structural lever for a verifier, not a containment primitive for the signing host.

The contributions are five; the four theorems among them are mechanized in a paper-local Lean model kept beside the paper, not in the substrate’s proof root (Section 9).

- A headline existence theorem: under any constitution with a non-empty required set and any body-admitting body, there exist two attestations sharing identical body bytes that discharge the admission predicate to opposite verdicts (`admission_predicate_separates_healthy_and_degraded_witnesses`). The witnesses are concrete: a healthy attestation listing one healthy record per required provider, and an empty attestation that fails coverage whenever the required set is non-empty.

- A partition-contingency biconditional placing the parent paper’s ladder mode at the proper-sublist relation between the attested-healthy providers and the required providers, with sublist witness paired with list inequality (`partition_contingency_mode_iff_degraded_subset`).
- A destructive-admission-witnesses-coverage projection: any admitted receipt, viewed through the admission predicate’s three-conjunct shape, carries a declared required-set and an attestation covering that set (`healthy_attestation_required_for_destructive_admission`).
- An amendment-improvement theorem: a receipt held in partition contingency under a prior constitution and re-admitted under an amended constitution carries a re-admission witness whose attestation is no longer in partition contingency under the amended substrate, on the same attestation bytes (`degraded_sensor_admission_requires_re_admission_witness`).
- A canonical-JSON encoding deployed against six host-snapshot emitter sites out of nineteen surveyed in a working admission kernel, with the remaining thirteen covered by a placeholder single-active-agent attestation that the constitution can reject by required-set declaration; the deployment instance retires the parent paper’s substrate-honest-by-assumption row as a structural matter rather than as operational discipline.

2 Background

Polity admission. A polity is a triple (T, C, K) : a closed receipt-admission scope T , a Merkle-rooted citizenship roster C , and a constitution K given as a finite list of predicates. Admission of a receipt r is the decidable Boolean

$$\text{polityAdmits}(P, r) = T(r) \wedge \text{allPredicates}(K, r).$$

A bilateral treaty admits r iff treaty scope, treaty constitution, and both polities’ admission relations accept. That equivalence is `treaty_admission_iff_predicate_intersection`, a bounded-model theorem in the substrate’s Lean proof root and its theorem inventory; the two theorems this paper borrows from the substrate are of that kind, and neither is a statement about the deployed Rust. The parent paper fixes the wire format: signed Ed25519 over canonical-JSON [31], DSSE envelopes binding bodies via subject digest [36], fail-closed on malformed envelopes and predicate-type mismatches.

The five-mode trust ladder. The parent substrate carries a finite ladder of admission modes, each a tag the constitution declares as the receipt family’s floor. Admission stability is monotone in the ladder: a receipt admitted at mode m is admitted at every mode below m , and treaty admission holds under any mode at or above the treaty’s declared floor `treaty_admission_stable_under_ladder_floor`, the second inventory theorem this paper borrows. The five modes, in ascending order of admission friction, are inherited from the parent construction [5]:

- *Observation.* The receipt is logged to the polity’s history without admission to any downstream effect. The ladder’s permissive floor; used for read-only attestations and trace records.
- *Guarded.* Admission requires policy-validation evidence and an ordinary signed audit trail, with no governance receipt; the mode for non-destructive escalation and advisory publication, and the highest mode at which a destructive action class is forbidden outright.

- *Receipt-backed.* Admission proceeds when a healthy substrate signs a body whose attestation covers the constitution’s required sensor set, with no reconciliation outstanding. The default mode for production traffic on a healthy kernel.
- *Partition-contingency.* Admission proceeds under a degraded substrate whose attested-healthy providers form a strict sublist of the constitution’s required set, with an explicit reconciliation obligation attached to the admitted receipt. The mode the present construction makes structurally decidable.
- *Maintenance.* Admission requires authenticated operator presence; the highest rung, covering operator review, evidence export, replay, and key rotation. Quorum-required is a consistency model attached to an action class, not a ladder mode.

A constitution may also *refuse* a receipt family outright, which corresponds to the empty admission relation rather than a ladder mode. The construction extends the partition-contingency row by making its trigger condition structurally decidable on a substrate-attested field, extends the receipt-backed row by binding admission to coverage of the constitution-required sensor set, and tightens quorum-required action classes by requiring every cosigner’s attestation to cover that set so a degraded cosigner cannot satisfy quorum from a clean body alone (§3).

The honest-substrate assumption. The parent construction is sound up to one limitation it names: the verifier-owned trust stores and the signing kernel are honest with respect to their own substrate state. A kernel that has lost a sensor at decision time can still produce a body that parses cleanly and signs cleanly. The bilateral predicate confirms that two kernels signed the same body and agreed on the same canonical bytes; it cannot distinguish two kernels that signed under complete observation from two kernels that signed under partial observation. The honest-substrate assumption is therefore not a separate cryptographic layer but a stipulation that runs through the entire admission predicate, occupying the position that the Coker principle of *trustworthy mechanism* holds in the formal-models-of-attestation literature [14].

Sensor state on real kernels. Real admission kernels are composed of multiple sensors: kernel callouts, network-flow verdicts, code-signature drift detectors, supply-chain runtime guards, behavioral telemetry. Each reports installed, active, healthy, degraded, and event-dropped state at runtime, and a kernel produces receipts using whichever are live at decision time. The receipts inherit, implicitly, the posture of the sensors that contributed to them. The kernel knows which sensors were live; the verifier does not. Lifting that knowledge into a signed attestation that travels with the receipt closes the gap structurally.

OS-level referents. The sensor categories have direct referents in production endpoint stacks. On Linux, the audit subsystem and the Integrity Measurement Architecture supply kernel callouts and code-signature evidence [18, 34]. On macOS, the Endpoint Security framework exposes a typed event stream covering process, file, and signing-identity activity [6]. On Windows, Event Tracing for Windows is the analogous high-volume provider channel that endpoint products consume [26]. The sensor-grounded admission construction sits at the admission layer above these provider channels rather than at the endpoint-detection product layer.

Property attestation as the lineage. The sensor-state attestation introduced in Section 3 joins a tradition. Load-time integrity measurement on a TCG root attests which code is running [34];

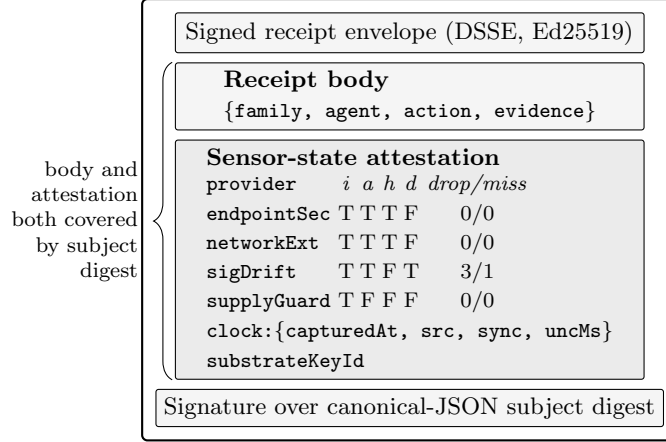


Figure 1: Sensor-state attestation traveling with every signed receipt. The substrate’s signing key covers both the body bytes and the per-sensor health record, so the verifier evaluates substrate posture as part of admission, not as a side channel. Flag column headings: *installed*, *active*, *healthy*, *degraded*.

semantic remote attestation lifts the attested object from *which code* to *which property of the code* [19]; the principles-of-remote-attestation program formalizes what an attestation must carry to be useful [14]. Trusted-execution-environment platforms (Intel TDX, AMD SEV-SNP, AWS Nitro, ARM CCA, NVIDIA Hopper Confidential Compute, Microsoft Azure Attestation) [11, 1, 32, 27] attest at a different layer: launch measurements rather than per-receipt sensing state. A TEE-rooted kernel that attests degraded sensing is structurally degraded under the admission predicate even if its TEE attestation is healthy. Section 8 returns to this lineage in depth.

3 The Sensor-State Substrate

The substrate gains one structural addition: a signed attestation of the kernel’s sensing posture, embedded in every receipt as a first-class field, canonicalized into the same subject digest as the receipt body. The attestation is not metadata. It is constitutional evidence at the same load-bearing rank as the receipt body, and the admission predicate is conditioned on it.

Sensor-state attestation. A sensor-state attestation is a finite list of provider records. Each record names a provider identifier, a provider kind tag (kernel callout, network-flow verdict, signature-drift detector, supply-chain runtime guard, behavioral telemetry, or a small fixed set of other tags), four Boolean state flags (*installed*, *active*, *healthy*, *degraded*), a list of degradation reason strings, a non-negative count of dropped events since the last attestation reset, and a non-negative count of deadline misses over the same window. The flags and counts are reported by the kernel’s own observability path, which is part of the kernel’s trusted code base; the attestation is a falsifiable signed claim about that path’s state at the captured timestamp. Figure 1 shows how the per-provider records sit beside the body inside the signed envelope.

The four flags are not redundant. A provider *installed* but not *active* is configured and quiescent. A provider *active* but not *healthy* is running but not producing trustworthy output (clock skew, missing entitlement, hung downstream channel). A provider *healthy* but *degraded* is producing output with known limitations recorded in the degradation-reason list. The four flags together

describe a provider’s posture at a resolution where the constitution can name which combinations admit which receipt families.

Clock state. The attestation also carries a clock record: the kernel-reported captured-at timestamp, the clock source identifier, a synchronized flag, and an optional millisecond uncertainty bound. It is in the same canonical-JSON object, covered by the same subject digest, and treated as a falsifiable claim, exactly like the provider flags. A constitution requiring *synchronized = true* and *uncertainty.ms* below a threshold can reject any receipt whose clock attestation falls outside it.

Required-set predicates. A constitution declares a required sensor set for each receipt family it admits. The required set is a finite list of provider identifiers and provider kinds; the admission predicate succeeds iff every required entry has a corresponding attestation entry that is *installed*, *active*, and *healthy*, with dropped-event and deadline-miss counts not exceeding the constitution’s declared per-family thresholds. The required set is a syntactic field of the constitution and travels with it across amendments. A constitutional amendment that adds a provider to a required set widens the admission predicate’s substrate dependency; an amendment that removes a provider must carry a backward-refinement witness over the predicate, as in the parent paper’s amendment cycle.

Falsifiable but not externally audited. The attestation is signed by the same key that signs the receipt body. A kernel that loses a sensor under an operational-discipline substrate can degrade silently; a kernel here must sign a false attestation, which an out-of-band sensor-coverage auditor can contradict by inspecting the kernel’s actual posture. The strengthening is strict. The paper does not specify the audit channel; it specifies the structural placement that makes the audit channel coherent.

TEE-rooted attestation-key separation. The substrate admits a deployment axis that replaces the single signing key with a hardware-rooted attestation-signing identity the workload cannot extract. Intel TDX binds quote signing to a hardware-rooted attestation-key path the TD cannot extract, with the TDREPORT and Quote flow specified by Intel [21]; AMD SEV-SNP signs under a Versioned Chip Endorsement Key the guest cannot control [2]; Apple’s Secure Enclave derives a User Identity Key inside its PKA hardware block, not readable by sepOS [6]; TPM 2.0 binds an Attestation Key under an Endorsement Key whose private half is sealed in TPM hardware. The pattern is uniform: a hardware-rooted key signs the sensor-state attestation, an application-controlled key signs the receipt body, and the verifier accepts the attestation only if its key is certified back to a platform endorsement key [9]. Threshold-Schnorr cosigning (FROST, MuSig2) is the cryptographic complement, producing a signature syntactically compatible with the present wire format.

Within-window discretization. The four flags discretize a state space that is, in reality, continuous over the decision window. A provider that delivers 99.7 percent of events with bursts of brief deadline misses is mapped to a single categorical record at attestation time; a provider that flaps at 10 Hz within a 100 ms decision is collapsed to whichever value the kernel’s observability path samples last. The schema treats the categorical state as the dominant condition across the decision window and treats within-window oscillation as captured by the drop and miss counts rather than by the flags. The continuous-suspicion alternative is the phi-accrual failure-detector

family, which derives a real-valued suspicion signal from heartbeat inter-arrival times [20]; the discretization threshold is a constitutional parameter and a continuous extension is named as future work [10].

Ladder reading. The five-mode trust ladder gains a structural definition for the partition-contingency mode. Once a placeholder for reconciliation evidence under expected partition, partition-contingency is now the case where the attestation shows the attested-healthy providers are a strict subset of the constitution-required providers. Receipt-backed and above require the attested-healthy set to cover the required set. Observation and guarded modes are below the destructive-admission floor and are decidable on the body alone; they do not consult the attestation. Admission of a quorum-required action class additionally requires that every cosigner’s attestation cover the required set; a degraded cosigner cannot satisfy quorum even if its receipt body matches.

The ladder is finite; the structural distinction between healthy and degraded is decidable; the partition-contingency mode is no longer a heuristic. The substrate carries one predicate that the parent paper named in the assumption ledger and not as a theorem.

4 Formal Model

The model adds a sensor-state attestation type, a sensor-attested receipt type, and an admission predicate conditioned on both fields to the parent paper’s predicate language. The construction is at the predicate layer; no cryptographic primitive is assumed.

Provider records and attestations. Let `ProviderKind` be a finite enumeration over a small fixed tag set. A provider record is a tuple

$$\rho = (id, kind, inst, act, healthy, deg, reasons, drop, miss),$$

with $id \in \text{ProviderId}$, $kind \in \text{ProviderKind}$, the four flags in $\{\perp, \top\}$, $reasons$ a finite list of strings, and $drop, miss \in \mathbb{N}$. A sensor attestation A is a finite list of provider records, together with a clock record $(\tau, src, sync, unc)$ as defined in Section 3.

The predicate `providerHealthy`(A, id) holds iff A contains a record with the given identifier whose installed, active, and healthy flags are all $\top$ and whose degraded flag is $\perp$, and is decidable on A .

Sensor-attested receipts. A sensor-attested receipt $\hat{r}$ is a pair (r, A) where r is a canonical receipt body and A is a sensor attestation. The canonical-JSON subject digest covers both fields jointly: two receipts $\hat{r}_1 = (r, A_1)$ and $\hat{r}_2 = (r, A_2)$ that share a body but differ on attestation have distinct subject digests, so the receiver cannot silently treat them as the same.

Constitution-required sets. A constitution K is a finite list of predicates over receipts as in the parent paper, additionally carrying a required-set map $\text{Req}_K : \text{ReceiptFamily} \rightarrow \text{Set}(\text{ProviderId} \times \text{ProviderKind})$ and per-family thresholds $\text{Drop}_K, \text{Miss}_K : \text{ReceiptFamily} \rightarrow \mathbb{N}$.

Admission under attested substrate. The sensor-attested admission predicate is

$$\begin{aligned} \text{admissibleUnderSensorState}(K, \hat{r}) = & \\ & \text{allPredicates}(K.\text{body}, r) \\ & \wedge \text{requiredSetCovered}(K, \hat{r}) \\ & \wedge \text{dropAndMissBounded}(K, \hat{r}), \end{aligned}$$

where `requiredSetCovered` holds iff every entry in Req_K for $\hat{r}$'s family corresponds to a `providerHealthy` entry in A , and `dropAndMissBounded` holds iff every provider in A has dropped-event count below Drop_K and deadline-miss count below Miss_K for the receipt's family. A provider marked both healthy and degraded does not satisfy `providerHealthy`. Both auxiliary predicates are decidable on A and K .

Headline theorem. For every constitution K whose required set Req_K for the body's family is non-empty, and for every body r that admits under K 's body predicates, there exist attestations A_h, A_d such that $\text{admissibleUnderSensorState}(K, (r, A_h)) = \top$ and $\text{admissibleUnderSensorState}(K, (r, A_d)) = \perp$.

The proof is a Σ -construction. It destructs the body-admission hypothesis to obtain a body r , instantiates the existential with two fixed attestations (a healthy attestation listing one healthy provider record per entry in Req_K , with installed, active, healthy all $\top$ and zero drop and miss counts; and an empty attestation), and discharges the admission predicate at each branch by rewriting under the family lookup and the supporting lemmas. The list-induction work lives inside the supporting lemmas: `requiredSetCovered.healthyWitness` lifts a per-entry healthy claim over Req_K via `List.all_eq_true`, and `not_requiredSetCovered.degradedWitness` case-splits on Req_K with the cons case discharged by `simp` and the nil case ruled out by the non-emptiness hypothesis. The headline itself does not induct over the provider list; it instantiates two fixed witnesses and rewrites. The empty list suffices as a degraded witness; the stronger separation that the predicate reliably rejects any short-falling attestation is the required-set-coverage projection (Theorem 3).

Worked example. Take K with required set on the response-execution family

$$\begin{aligned} \text{Req}_K = \{ & (\text{endpointSecurity}, \text{kernelCallout}), \\ & (\text{networkExtension}, \text{networkFlow}) \}, \end{aligned}$$

$\text{Drop}_K = \text{Miss}_K = 0$, and let r be a canonical response-execution body that body-admits under K . The healthy witness pairs r with $A_h = [\rho_{\text{ep}}, \rho_{\text{net}}]$, each ρ naming its provider with all four flags installed, active, healthy, not-degraded and counts $(0, 0)$. The degraded witness pairs the identical r with $A_d = []$. Both receipts share body bytes; their subject digests differ because the digest covers both fields. The admission predicate discharges to $\text{admissibleUnderSensorState}(K, \hat{r}_h) = \top$ and $\text{admissibleUnderSensorState}(K, \hat{r}_d) = \perp$: the required-set check matches both Req_K entries on A_h and fails `endpointSecurity` on A_d . The structural distinction lives in the attestation, not the body.

Ladder structuring. The partition-contingency mode is

$$\begin{aligned} \text{partitionContingencyMode}(K, \hat{r}) \Leftrightarrow & \\ \text{attestedHealthy}(K, \hat{r}) \sqsubset \text{Req}_K(\hat{r}.\text{family}), & \end{aligned}$$

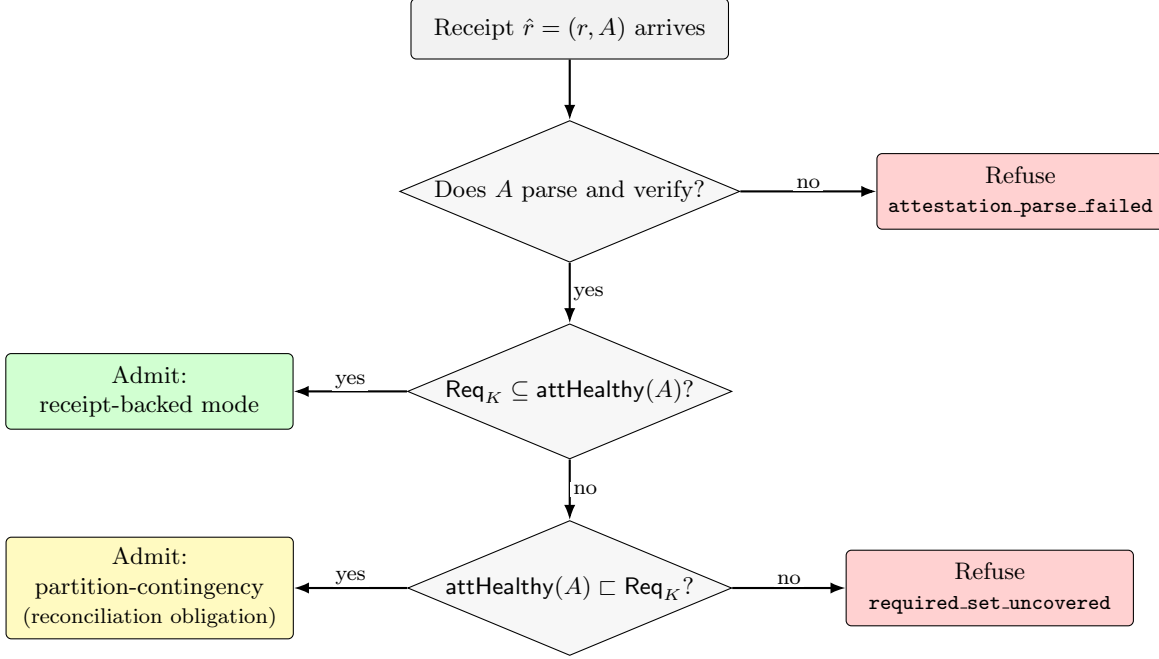


Figure 2: Admission decision tree. The attestation field drives a three-way verdict: receipt-backed admission when the healthy sensor set covers the constitution’s required set; partition-contingency admission with a reconciliation obligation when the healthy set is a strict sublist of the required set; refusal otherwise. The decision is structurally decidable on the attestation field; Theorem 1 names the witness pair, Theorem 2 names the strict-sublist biconditional.

where $\sqsubset$ denotes the strict-sublist relation (sublist witness paired with list inequality) and `attestedHealthy` is the order-preserving subsequence of `ReqK` entries that the attestation asserts healthy. The biconditional is *partition_contingency_mode_iff_degraded_subset*. The forward direction extracts the sublist witness from `attestedHealthy = filter(·, ReqK)` and establishes list-distinctness from the strict length inequality. The backward direction recovers the strict length inequality via `Sublist.length_le` together with `Sublist.eq_of_length`. Both rely on the structural lemma that filtering preserves the sublist relation.

The two cases the biconditional partitions, together with the rejection branch when neither the covering nor the strict-sublist condition holds, render as the three-way decision diagram in Figure 2: the constitution’s required set drives a single traversal of the attestation, and the verdict is one of receipt-backed admission, partition-contingency admission with a reconciliation obligation, or refusal with a typed code.

The destructive-admission floor is conditioned on a projection of the admission predicate’s three-conjunct shape. The supporting theorem *healthy_attestation_required_for_destructive_admission* extracts the required-set coverage Boolean from any admitted receipt via a `Bool.and_eq_true` decomposition. The destructive-family discriminator appears in the signature for prose alignment with the floor’s intended invariant but does not appear in the proof body; the projection holds for any admitted receipt, and admission of a quorum-required action class rejects any cosigner whose attestation falls short of the required set.

Amendment re-admission. The supporting theorem *degraded_sensor_admission_requires_re_admission_witness* states that given `lookupRequiredSet( $K, \hat{r}.family$ ) = declprev`,

$\text{partitionContingencyMode}(\text{decl}_{prev}, A) = \top$ and $\text{admissibleUnderSensorState}(K', \hat{r}) = \top$, there is a re-admission witness over the same attestation bytes A whose existential conclusion carries the prior lookup $\text{lookupRequiredSet}(K, \hat{r}.family) = \text{decl}_{prev}$ alongside the amended lookup $\text{lookupRequiredSet}(K', \hat{r}.family) = \text{decl}_{next}$, the prior mode $\text{partitionContingencyMode}(\text{decl}_{prev}, A) = \top$, and the new mode $\text{partitionContingencyMode}(\text{decl}_{next}, A) = \perp$. Carrying the prior lookup in the conclusion ties the improvement to both constitutions rather than to the amended one alone. The amendment narrows the substrate-required set far enough that what fell short of $\text{Req}_K(\hat{r}.family)$ now covers $\text{Req}_{K'}(\hat{r}.family)$ in full: a structural improvement, not a relabel.

Threat model. The parent paper’s adversary controls the agent process, malformed evidence, and stale treaty references; cannot break the listed cryptographic primitives; and is honest-by-assumption with respect to its own substrate state. The threat model here strictly strengthens this last clause: a kernel that silently lost a sensor mid-decision and produced a body that parses cleanly is now also signing an attestation that asserts the lost sensor is healthy. The silent-degradation path becomes a signed-false-attestation path, falsifiable by any out-of-band sensor-coverage auditor; the parent admission predicate, conditioned only on body fields, could not state that audit obligation structurally. Attestation and body are signed by the same kernel key in the construction here; a separate attestation key, rooted in a TEE platform’s quote-signing identity, would make the strengthening cryptographic in addition to structural and is named as an extension axis.

The model does not introduce the audit channel; it introduces the structural placement that makes one coherent. A polity that admits a receipt without evaluating its attestation has chosen to skip a check the substrate makes available, and a polity that admits under a degraded attestation has chosen the partition-contingency mode rather than the receipt-backed mode. Both choices are recorded in the receiver’s history as constitutional events.

5 Implementation

The admission kernel reuses the parent paper’s Rust runtime; the structural addition is the canonical-JSON encoding of the sensor attestation and the predicate evaluator that consumes it.

Attestation encoding. The attestation is a canonical-JSON object with a fixed key set: a `providers` array and a `clock` record. Each provider entry has the eleven fields named in Section 4, encoded in camelCase. The provider-kind tag is a string with a finite enumeration: `kernelCallout`, `networkFlow`, `signatureDrift`, `supplyChainGuard`, `behavioralTelemetry`, `agentApi`, `honeyToken`, `other`. The clock record has the four fields named in Section 3. The canonical-JSON encoding follows RFC 8785 with the same field-ordering and number-canonicalization discipline as the receipt body [31]. The attestation digest is the SHA-256 hash of the canonical bytes; the DSSE subject digest is over the concatenation of body bytes and attestation bytes under a fixed delimiter.

Required-set declaration. The constitution carries the required-set map and per-family thresholds in a separate canonical-JSON block adjacent to the predicate list. The block is parsed at constitution load time and amends only through the backward-refinement witness and re-admission obligation of Section 4. A constitution whose required-set block fails to parse rejects every receipt under the fail-closed discipline in the admission hook `validateTreatyScope`.

Predicate evaluator. The evaluator adds two checks to the parent paper’s path. The required-set check consults the attestation’s provider list against the receipt family’s required entries; each

required entry must match a provider entry with installed, active, and healthy flags all set and degraded unset. The drop-and-miss check verifies each provider’s dropped-event and deadline-miss counts are within per-family thresholds. The evaluator returns one of four outcomes: receipt-backed admission, partition-contingency admission, observation-mode admission, or denial. The parent paper’s mode-floor check composes: a treaty requiring receipt-backed admission rejects a partition-contingency outcome.

Failure modes. The evaluator fails closed on every parsing error in the attestation block, every malformed provider record, every signature mismatch between the attestation digest and the DSSE subject digest, and every constitution whose required-set declaration is malformed. Failure paths emit a denial receipt with a typed denial code: `attestation\_parse\_failed`, `required\_set\_uncovered`, `drop\_count\_exceeded`, `deadline\_miss\_exceeded`, or `attestation\_subject\_digest\_mismatch`. The denial receipt carries an attestation of the kernel that produced it, closing the self-referential gap of denials issued under a degraded substrate. An attestation with a syntactically valid but empty `providers` array passes the parser and discharges to `required\_set\_uncovered` whenever the constitution’s required set is non-empty; the empty-providers attestation is therefore wire-producible by construction, and the headline theorem’s degraded witness is reachable through the predicate-level denial path.

Emitter populations. A receipt-emitting deployment carries one of two attestation populations. The first derives the provider list from a real host snapshot through `endpoint\_sensor\_state\_from\_macos\_host`; six emitter sites consume it, covering response-execution receipts and the sensor-state receipt itself. The second is a structural constant, `EndpointSensorState::single\_active\_agent("agent-api")`, a one-record attestation marked installed, active, and healthy. Thirteen sites consume the constant, covering policy-decision, telemetry-privacy, graph-slice, simulation, deception, plan, rollback, and acknowledgement receipts. The hash-bound rows are coherent on the constant by construction, so the validator admits both under identical rules; what differs is provenance, and a verifier distinguishes them syntactically from the single-provider record and the literal constructor. The empirical claim covers the first class; the second class is a placeholder population whose admission under the same predicate is a structural limitation, recorded as such in Section 9.

6 Evaluation

The empirical claim is bounded to the substrate’s verifier-side discipline on a sensor-state receipt. Absent measurements are marked rather than estimated.

Verifier-side rejection on attestation mutation. The validator at `require\_sensor\_state\_evidence` hash-binds six evidence rows: `providerCount`, `activeProviderCount`, `healthyProviderCount`, `degradedProviderCount`, `providerIds`, `sensorStateHash`. Each must match a re-hash over the receipt’s `sensor\_state` field under a fail-closed rule-id assertion. A companion check, `require\_provider\_degradation\_consistency`, rejects any record whose flags and degradation-reason list contradict the `degraded` flag.

The load-bearing measurement is a verifier-side mutation test, `endpoint\_sensor\_state\_receipt\_binds\_provider\_health`. It signs a two-provider receipt (healthy `agentApi`; degraded `macos.endpoint\_security` with `dropped\_event\_count=2`, `deadline\_miss\_count=1`, missing full-disk access), then perturbs it one mutation at a time and asserts `validate()` returns

Err. Mutations cover tampered `providerCount`, `providerIds`, `activeProviderCount`, and `sensorStateHash` rows; a relabeled provider identifier; an inconsistent `degraded` flag; missing or future-dated `last\_seen` on an active provider; and a cleared or blank degradation-reason list. Every mutation is denied with a substring-matched error. `healthyProviderCount` and `degradedProviderCount` are exercised transitively, since the consistency check rejects flag-field perturbations before those rows are reached; a coordinated mutation that bypasses the consistency check to perturb them directly is named below as withheld.

Receipt-emitter survey. The macOS host snapshot is built by `endpoint\_sensor\_state\_from\_macos\_host` in `api\_server`. Six emitter sites consume it, covering response-execution receipts and the sensor-state receipt itself. Thirteen further sites use a placeholder, `EndpointSensorState::single\_active\_agent("agent-api")`: one record, healthy, no degradation. The placeholder is a structural constant; the validator’s hash-bound rows are coherent on it by construction. The empirical claim covers the strict subset of receipts emitted at the six host-snapshot sites.

Verification cost. The validator re-hashes six evidence rows and the canonical-JSON encoding of `sensor\_state` via `endpoint\_sensor\_state\_content\_hash`. The work is $O(\text{provider count})$ under SHA-256. No latency harness exists in the substrate’s bench surface; a Criterion bench over the sign and validate paths is withheld. The canonical-JSON byte size is bounded by provider count: roughly 200–400 bytes for the placeholder, 600–1200 bytes for a three-provider host snapshot. Receipt-ledger storage grows linearly in provider count.

What is not measured. The macOS `EndpointSecurity` system extension shipped with the substrate sources events from an in-memory recorder rather than calling `es\_new\_client` or `es\_subscribe`. The host-snapshot path’s ES provider record is therefore real in shape but synthetic in content: runtime flags, install state, drop and miss counts, and full-disk-access signal are recorder-supplied, not derived from a live ES client. The sensor input layer the empirical claim covers is the network-extension filter (real on the egress reload path), the package-manager hooks, and the tool-preflight surface; ES-derived telemetry is out of scope.

A deployment-rate partition-contingency number is not reported: aggregating host-snapshot and placeholder populations would conflate a host posture with a structural constant. A false-attestation rate is not reported (Section 9): no out-of-band sensor-coverage auditor is part of the substrate. The Criterion latency bench, and a direct mutation across `healthyProviderCount` and `degradedProviderCount` on an attestation that bypasses the consistency check, are withheld; the structural placement of both is well-defined.

7 Discussion

The structural argument. The contribution is not a new cryptographic primitive. The contribution is a placement choice and the formal consequences of that choice. The kernel knows its own sensors are degraded; without an attested sensor-state record, that knowledge is a local, unobservable fact; with one, it is a signed, falsifiable claim that travels with the receipt. The verifier can no longer be honest-substrate-by-assumption; the verifier must evaluate the substrate’s attestation or explicitly choose to skip it. Both choices are constitutional events.

What sensor-grounded admission reveals that the trust-store-honest model hides. The structural distinction is observability, not detection. Under the parent paper’s trust-store-honest row, a receipt whose body parses cleanly and signs cleanly is admitted at the receipt-backed mode, and a kernel that has silently lost a sensor cannot be told apart on the wire from one that has not. Under the sensor-grounded construction, the same body bytes travel with an attestation that names which providers were healthy at decision time, so a receipt produced under partial sensing is admitted at partition-contingency rather than receipt-backed, with an explicit reconciliation obligation attached. The reviewer reading the receipt sees the degradation as a structural artifact of admission; under the prior model, the reviewer would not have seen it at all. The cross-substrate effect is sharper still: a bilateral treaty whose required set is the union of two polities’ required sets rejects a receipt whose attestation is degraded on a sensor that only one polity required, where the prior model admitted it.

Why this is not heuristic substitution. A reviewer may object that this is a renamed heuristic: “if the sensor is degraded, treat the receipt as suspect” is the same advice an EDR vendor’s runbook would give. The objection misses the structural distinction. A heuristic is a recommendation to a human operator; a predicate is a Boolean over the receipt’s substrate-attested fields, decidable by the verifier without operator judgment, and connected to the parent paper’s amendment cycle by the re-admission theorem. The structural distinction shows up in three places: (1) the partition-contingency mode is no longer a placeholder but a decidable predicate on the attestation; (2) the destructive-admission floor is type-conditioned on healthy attestation rather than gated by ladder rank as a convention; (3) the amendment cycle is conditioned on a re-admission witness, not on operator confirmation.

Composition with the parent paper’s ladder. The five-mode ladder is now driven by structural data rather than by operator discretion. Receipt-backed and partition-contingency, in particular, were adjacent rungs whose distinction the parent paper carried as a row in the assumption ledger; the sensor-grounded construction promotes that row to a decidable predicate on the attestation field, and the choice between the two modes is now a function of the receipt itself rather than an operator’s reading of recent kernel health. The composition is structural rather than substitutive: the headline theorem composes with the parent paper’s bilateral-treaty admission, the ladder-floor stability theorem composes with the constitution-required-set discipline, and the amendment-refinement theorem gains a paired re-admission obligation. A treaty between two polities admits a receipt iff both polities’ admission predicates accept; with the sensor-attested extension, both predicates also require coverage of their own required sets on the receipt’s attestation. The cross-polity admission is therefore strictly tighter than under the parent paper’s predicate alone, with no additional cryptographic machinery beyond the body-and-attestation digest binding.

Boundary against TEE-compromise. The construction inherits the TEE root-of-trust assumption from the parent substrate. Where the TEE root holds, the construction strictly improves observability over the trust-store-honest model: a kernel that loses a sensor must sign a false attestation, which is a falsifiable claim that an out-of-band sensor-coverage auditor can contradict, and the bilateral-treaty composition tightens the cross-polity admission predicate. Where the TEE root fails, the construction collapses to single-key signing as the parent paper assumed. The collapse is named structurally in Section 9 and decomposes by attack family: microarchitectural side channels (TDX-Down, Downfall, Plundervolt) and voltage-glitching faults push the residual one layer down rather than to zero. The construction does not promise containment when the TEE

root is breached; it promises that, where the root holds, sensor coverage is a decidable property of the receipt rather than an operator-discretion field.

The downstream claim the placement choice unlocks. The structural placement is the precondition for any coherent sensor-coverage auditor whose disagreement with the kernel is itself a signed artifact: an arrangement in which sensor-state never appears on the wire cannot support such an auditor at all. The auditor is named as the structural complement of this construction, not specified here; what travels on the wire after this construction is the falsifiable claim that lets the auditor exist. Symmetrically, the partition-contingency mode is now a treaty primitive: a bilateral treaty whose required set is the union of two polities’ required sets is decidable across an organizational trust boundary without either polity adopting the other’s operator-discretion vocabulary. Both downstream artifacts are out of scope for the present construction and in-scope for the structural placement that makes them coherent.

8 Related Work

Property attestation. Sailer et al. gave the canonical Linux-side load-time attestation through the TCG Integrity Measurement Architecture, chaining hash measurements of every loaded executable from firmware into the running kernel [34]. Halder, Chandra, and Franz lifted the attested quantity from a measurement chain to a property: a verifier accepts an attestation iff the attested system has the property the verifier requires, irrespective of which binaries produced it [19]. Sadeghi and Stübke formulated the same shift independently [33]. Coker et al. gave the formal-models account: attestation is fresh, comprehensive, constrained-disclosure, semantically explicit, and rooted in a trustworthy mechanism [15]; IETF RATS codifies the same shape as a claims model the verifier evaluates against an appraisal policy [9].

Trusted sensors. Saroiu and Wolman gave the founding construction: a hardware sensor signs its own reading so a downstream consumer can verify the value [35]. Liu, Saroiu, Wolman, and Raj generalized the primitive into OS abstractions for software consumers of trusted sensor data [24]. Asokan et al. extended the predicate-on-attested-state shape to collective embedded-device attestation [8]. Sensor-grounded admission inherits both branches and lifts them from the input layer (a single sensor signs a single reading) to the substrate layer (a kernel signs a federation of sensors’ health postures).

TEE attestation and what its wire formats cannot express. TEE platforms produce attestation evidence whose schema is fixed by their wire formats. Intel TDX exports a build-time MRTD digest, four runtime measurement registers (RTMR0-3), and a 64-byte REPORTDATA slot through the TDX Quote v4 / v5 structure [21]. AMD SEV-SNP issues an ATTESTATION_REPORT with a launch-only MEASUREMENT field and an uninterpreted 64-byte REPORT_DATA slot [2]. AWS Nitro Enclaves emit a COSE_Sign1 document with PCR0-PCR8 slots and a 512-byte user_data field [4]. Apple Private Cloud Compute carries a SealedHashLedger of secure-config and cryptex measurements bound to a public transparency log [7]. Arm CCA’s realm-token claims expose RIM and REM extension chains under the RATS Entity Attestation Token [16]. NVIDIA Hopper Confidential Compute, deployed on H100 and H200 GPUs, attests per-device measurements through SPDMM-tunnelled GPU attestation reports binding the on-chip firmware version, secure-boot state, and a workload-supplied nonce; the wire format extends the device-attestation surface to GPU accelerators that increasingly host agentic-AI workloads [30]. Google Cloud Confidential Computing

layers SEV-SNP and TDX guests under a Confidential Space attestation token, signed by Google’s Attestation Verifier, that carries TEE measurements and a workload identity claim consumed by downstream verifiers [17]. The CNCF Confidential Containers project composes a Kubernetes-native trusted-execution path with a Remote Attestation Service whose Reference Value Provider supplies policy-side acceptance rules over the per-pod TEE evidence [13]. Microsoft Pluton, the security-processor architecture deployed across Windows-class PC silicon, anchors a hardware root that the platform’s attestation chain certifies up to Microsoft Azure Attestation [25, 27]. No surveyed wire format expresses per-sensor drop or deadline-miss counts, nor distinguishes “sensor never installed” from “sensor degraded within the window.” Runtime measurement registers extend SHA-384 hash chains over post-launch events but collapse a degraded provider and an uninstalled provider onto indistinguishable extension chains. The substrate retains compositional independence: a TEE-rooted kernel whose sensor attestation reports degraded providers remains degraded under the admission predicate, and the two attestations compose at distinct layers rather than substituting for each other. The construction is substrate-agnostic with respect to the underlying attestation system; it consumes any of the surveyed wire formats as the carrier for its hardware-rooted attestation-signing key, and overlays a sensor-coverage claim that the underlying format does not itself express.

Kernel observability and the eBPF lineage. The Linux audit subsystem is the canonical prior art for “the OS counts what it dropped”: a kernel-side queue capped by `backlog_limit`, with lost-event statistics reported through `auditctl -s` and a configurable failure mode on overflow [18]. The eBPF-era replacement lineage tightens the same shape. Sekar, Kimm, and Aich’s *eAudit* motivates itself by the observation that legacy audit pipelines lose a majority of events under sustained workloads, and is built on eBPF for high-throughput capture [37]. Falco decomposes kernel-side losses into buffer-pressure, page-fault, and verifier-condition counters exported to userspace [38]; Cilium Tetragon ships a parallel set under the same channel [12]. None of these binds the kernel-coverage attestation to a polity-constitution admission predicate; that placement is the structural delta over the kernel-observability layer.

Aerospace redundancy. Aerospace-grade sensor-failure handling solves disagreement through hardware redundancy and median voting rather than cryptographic evidence: the Boeing 777 Primary Flight Computer used triple-triple redundant lanes with mid-value-select voting [40], and ARINC 653 partitions reaction policy across application, partition, and module fault levels [3]. TMR voting produces a single output with no signed downstream-consumable record of the failure event and no admission boundary at which a downstream polity evaluates the failure as a predicate; the two layers compose, but cryptographic attestation of sensor coverage is novel relative to this lineage.

Endpoint detection and response. The substrate’s provider categories (kernel callouts, network-flow verdicts, code-signature drift detectors, behavioral telemetry) correspond to the data sources EDR systems consume on Linux through the audit subsystem and on macOS through the Endpoint Security framework [6]; code-signature drift is the data type produced when an installed binary’s signature chain diverges from a baseline at scale [23]. EDR consumes sensor data and emits human-readable alerts at the workstation tier; sensor-grounded admission consumes a structurally identical inventory but emits a signed canonical attestation that a polity admission predicate evaluates at the receipt boundary.

Parent admission substrate. The construction extends the predicate language and admission relation of the programmable-sovereignty substrate [5]. The parent’s theorems `treaty\_admission\_iff\_predicate\_intersection` and `treaty\_admission\_stable\_under\_ladder\_floor` compose with the sensor-attested admission predicate without modification; `amendment\_admissible\_iff\_backward\_refinement` gains a paired re-admission obligation, as Section 4 establishes. The sensor-attestation construction is a fresh constitution rather than an amendment of the parent body-only constitution; the backward-refinement obligation applies to amendments within the new constitution, not to its introduction.

9 Limitations

The contribution is a structural placement choice and the formal predicate that follows from it. The limitations are correspondingly structural.

Substrate-honest assumption is strictly strengthened, not eliminated. The parent paper’s assumption ledger named substrate honesty as a row of operational discipline. The substrate here strictly strengthens that row: a kernel that loses a sensor must now sign a false attestation, which is a falsifiable signed statement rather than a silent operational fact. An out-of-band sensor-coverage auditor whose disagreement with the kernel’s attestation is itself a signed artifact is the precondition for catching the lie; the auditor is named as the structural complement of this work and is not specified here. The key-isolation residual that lets a compromised kernel sign coherent false attestations is recorded separately below.

Required-set expressiveness. The required-set map is a finite list of provider identifiers and provider kinds per receipt family. It does not express per-tenant required sets, time-windowed thresholds (the threshold should ratchet down during specific windows), or conditional requirements (a network-flow sensor is required only when the action class is network egress). These are straightforward extensions in the same syntactic family, admitted as future extensions of the constitution’s syntax.

Drop-and-miss thresholds are not action-specific. The drop-and-miss thresholds are declared per receipt family rather than per action within a family. A constitution that wants strict drop-rate control on destructive actions and permissive control on observation actions must split the receipt family into a finer taxonomy. The receipt-family taxonomy is itself a structural primitive whose finer subdivision is a separate extension to the substrate.

Within-window flapping. The schema assumes the categorical sensor state is the dominant state across the decision window, and uses that assumption to discretize each provider into the four flags (installed, active, healthy, degraded). Within-window flapping is out of scope: a sensor that toggles healthy and degraded N times in the decision window is attested by whichever state dominated, with the others as collateral noise captured only by the drop and miss counts. The residual risk is that a constitution that needs to reject high-frequency oscillation has no schema field for transition rate and must subdivide the receipt family; the phi-accrual continuous-suspicion extension cited in Section 3 is named as the future-work alternative [20, 10].

Clock attestation is not separately validated. The clock record is a signed claim, not verified against an external time source. A constitution can declare an uncertainty bound and reject receipts that exceed it; a kernel that lies about its clock signs a false claim. The discipline matches the sensor-attestation discipline, and an external time-coverage auditor is its structural complement.

Attestation-key isolation. The strengthening of the construction rests on operational discipline rather than on cryptographic isolation of the body-signing and attestation-signing roles, which share one key. An endpoint that controls the body-signing key also controls the attestation; the sensor-state predicate then collapses to a structural-only claim with no additional cryptographic guarantee beyond what the body’s signature already supplies. Hardware-root-of-trust separation (Intel TDX TDREPORT path [21], AMD SEV-SNP VCEK [2], Apple SEP UIK [6], TPM 2.0 AK/EK) closes the gap structurally; threshold-Schnorr cosigning (FROST, MuSig2) closes it cryptographically with a signature output syntactically compatible with the single-key wire format. The RFC 9334 Attester / Target distinction is the standards-anchor for both closures [9].

TEE compromise reduces the construction to single-key signing. A hardware-root-of-trust closure for the attestation-key-isolation row pushes the residual one layer down rather than to zero. The TEE substrate that holds the attestation key has its own attack surface: voltage-glitching attacks against SGX have extracted enclave secrets [29]; transient-execution attacks have read keys through speculative side channels [39]; gather-data-sampling has lifted register state across boundaries on recent Intel parts [28]; Rowhammer-class fault injection has corrupted memory across privilege domains on commodity DRAM [22]. A TEE-rooted attestation key extracted by any of these paths reduces the construction’s structural strengthening to single-key signing for receipts produced during the compromise window: the attestation signature continues to verify under the verifier’s trust-store policy while no longer reflecting the kernel’s sensing posture. The construction does not defeat these attacks; it inherits the TEE root-of-trust assumption and is no stronger than the underlying TEE substrate. Hardware-enforced multi-key signing (a sensing-side attestation key distinct from the body-signing key, held in a separate cryptoprocessor), per-receipt attestation-key rotation, and escrow-backed recovery from a compromise window are the candidate mitigations whose structural fit with the sensor-grounded predicate is named as future work.

Cosignature party-independence. A bilateral DSSE envelope and the sensor-state attestation are separable cryptographic outputs only when their signers are independent principals: a single actor controlling both keys collapses the bilateral primitive to one-of-one signing, regardless of envelope shape. The construction here treats cosigner attestations for quorum-required action classes (Section 4) as joint authorization for cross-organizational admission; the residual is that a single-actor deployment whose body-signing and attestation-signing keys both rest with one principal recovers no cryptographic strengthening over single-key signing, only the structural strengthening of a falsifiable claim. The parent paper’s assumption ledger names this collapse; the present construction inherits the residual without modification.

Cross-polity required sets. A bilateral treaty’s required set is the union of both polities’ required sets for the receipt’s family. The construction admits this as the safe default; alternative reductions (intersection, weighted-quorum) are syntactic choices the constitution can declare. The intersection reduction in particular is dangerous: a treaty whose required set is the intersection of two polities’ requirements admits a receipt whose attestation is degraded on a sensor required by

only one polity, which retreats from the structural strengthening the construction contributes. The recommendation is to use the union as the structural default.

Theorem coverage. The headline theorem and the three supporting theorems are stated in Section 4 and mechanized in a paper-local Lean model. All four compile cleanly with no `sorry`, and `#print axioms` reports only the standard kernel axioms (`propext`, `Classical.choice`, `Quot.sound`). Several supporting lemmas (`providerHealthy_in_healthyWitness`, `requiredSetCovered_healthyWitness`, `dropAndMissBounded_healthyWitness`) resolve via straightforward unfolding rather than deep induction; the load-bearing list-induction work lives in Theorem 2’s proper-sublist biconditional, which composes `List.filter_sublist`, `Sublist.length_le`, and `Sublist.eq_of_length`. Theorem 3’s `destructiveAdmissionFamily` hypothesis is currently inert: the conclusion holds for any admitted receipt, and the binder is retained as a structural hook for a strengthened statement. Theorem 4 now carries the prior required-set lookup into the existential conclusion, so the structural-improvement witness is explicitly tied to both the prior constitution and the amended constitution. The four theorems are mechanized in the paper-local file `lean/SensorGroundedAdmission.lean`, which is checked against the substrate’s Lean root but is not part of it and is not in the substrate’s theorem inventory; keeping the file elaborating under refactoring or proof strengthening is a paper-side obligation, as in the parent paper’s ledger.

Operational observability gap not closed. The parent paper’s operational-observability row described the receipt log as the audit channel. The substrate here does not change that: it produces canonical statements but does not specify the live-observability protocol that downstream dashboards would consume.

Empirical chapter scope. The empirical chapter cites one substrate; cross-substrate corroboration is out of scope. A second substrate producing sensor-attested receipts under the schema of Section 3 is the structural complement that retires the single-substrate evidence basis.

Schema evolution inherited from parent. The parent paper carries a schema-evolution row covering version migration across vendor kernels with non-uniform schema upgrades. The sensor-attestation construction adds new wire fields (provider records, clock records) under the parent’s canonical-JSON discipline; it inherits the parent’s schema-evolution obligation without adding a sensor-grounded-specific extension, and a versioned-predicate compatibility profile remains the prerequisite for cross-kernel sensor-attested admission.

10 Conclusion

The polity admission relation in the parent substrate was sound up to one assumption: the kernel was honest with respect to its own sensing posture. The assumption was named, but it was carried as operational discipline rather than as a substrate-level predicate. This paper closes the gap by lifting the kernel’s sensing posture into a signed attestation embedded in every receipt, by conditioning the admission predicate on the attestation, and by giving the parent paper’s five-mode trust ladder a structural definition for the partition-contingency mode.

The contribution is a placement choice and the formal consequences of that choice. Four theorems, mechanized in a paper-local Lean model beside the paper, capture the consequences. The headline existence theorem (`admission_predicate_separates_healthy_and_degraded_w`

`itnesses`) constructs two attestations sharing the same body bytes that discharge admission to opposite verdicts under one constitution; the proof is a Σ -construction over a healthy attestation listing one healthy record per required provider and an empty attestation that fails coverage. The partition-contingency biconditional (`partition_contingency_mode_iff_degraded_subset`) places the parent paper’s ladder mode at a proper-sublist relation between the attested-healthy providers and the required providers, with `Sublist.length_le` and `Sublist.eq_of_length` carrying both directions. The admission-witnesses-coverage projection (`healthy_attestation_required_for_destructive_admission`) extracts required-set coverage from the admission predicate’s three-conjunct shape; any admitted receipt, viewed through that shape, has a declared required set and an attestation covering it. The amendment-improvement theorem (`degraded_sensor_admission_requires_re_admission_witness`) establishes that a re-admitted partition-contingency receipt is no longer in partition contingency under the amended substrate, on the same attestation bytes. The implementation instance covers six host-snapshot emitter sites out of nineteen surveyed in a working admission kernel, with the remaining thirteen attested by a placeholder single-active-agent record that a constitution can reject by required-set declaration; the substrate-honest-by-assumption row is retired as a structural matter rather than as operational discipline.

A kernel that lies about its sensors now signs a false claim. The substrate does not detect the lie internally; the lie becomes detectable only when an out-of-band sensor-coverage auditor disagrees with the kernel’s attestation, and that disagreement is itself a signed artifact. The substrate gains the precondition for that audit channel; the audit channel itself is the structural complement and is named as future work. The strict strengthening is conditional on the auditor’s existence; the substrate-side commitment is that a kernel that lies about its sensors is now signing a falsifiable claim rather than emitting a silent one.

The result is one step in a larger program. The admission predicate and the bounded treaty model this paper builds on are carried in the substrate’s Lean proof root; the polity vocabulary is this paper’s own. The present paper conditions the admission predicate on a substrate-attested field. The natural continuation is the same discipline applied to other substrate dependencies: a per-tenant required-set predicate, a time-windowed threshold language, a behavioral-attestation primitive for model-bound substrates. Each is a placement choice that lifts an operational assumption into a falsifiable signed statement; each retires a row from the assumption ledger.

References

- [1] Advanced Micro Devices. AMD SEV-SNP: Strengthening VM isolation with integrity protection and more. Technical report, Advanced Micro Devices, Santa Clara, CA, USA, 2020.
- [2] Advanced Micro Devices. SEV secure nested paging firmware ABI specification. Technical Report Publication 56860, Revision 1.58, Advanced Micro Devices, Santa Clara, CA, USA, May 2025.
- [3] Aeronautical Radio, Incorporated. ARINC specification 653: Avionics application software standard interface, part 1, required services. ARINC AECC Standard, Annapolis, MD, USA, 2015. Includes Part 1 Health Monitoring services.
- [4] Amazon Web Services. aws-nitro-enclaves-nsm-api attestation process. AWS Nitro Enclaves open-source documentation, 2024.
- [5] Anonymous. Receiver-Owned Bilateral Admission for Cross-Organization Agent Tool Calls. Working paper, 2026. Substrate paper extended by the present construction.

- [6] Apple Inc. Apple platform security. Apple Technical Documentation, 2024.
- [7] Apple Inc. security-pcc: CloudAttestation Proto AttestationBundle. Apple open-source repository, October 2024.
- [8] N. Asokan, Ferdinand Brasser, Ahmad Ibrahim, Ahmad-Reza Sadeghi, Matthias Schunter, Gene Tsudik, and Christian Wachsmann. SEDA: Scalable embedded device attestation. In *Proceedings of the 22nd ACM Conference on Computer and Communications Security (CCS)*, pages 964–975, New York, NY, USA, 2015. ACM.
- [9] Henk Birkholz, Dave Thaler, Michael Richardson, Ned Smith, and Wei Pan. Remote ATtestation ProcedureS (RATS) architecture. RFC 9334, January 2023.
- [10] Alvaro A. Cardenas, Saurabh Amin, and Shankar Sastry. Research challenges for the security of control systems. In *Proceedings of the 3rd USENIX Workshop on Hot Topics in Security (HotSec)*, Berkeley, CA, USA, 2008. USENIX Association.
- [11] Pau-Chen Cheng, Wojciech Ozga, Enriquillo Valdez, Salman Ahmed, Zhongshu Gu, Hani Jamjoom, Hubertus Franke, and James Bottomley. Intel TDX demystified: A top-down approach. *ACM Computing Surveys*, 56(9):238:1–238:33, 2024.
- [12] Cilium Authors. Cilium Tetragon: eBPF-based security observability and runtime enforcement. Cilium project documentation, accessed 2026-05, 2025.
- [13] Cloud Native Computing Foundation. CNCF confidential containers: Kata-rooted trusted execution with remote attestation service. CNCF Confidential Containers project documentation, 2024.
- [14] George Coker, Joshua Guttman, Peter Loscocco, Amy Herzog, Jonathan Millen, Brian O’Hanlon, John Ramsdell, Ariel Segall, Justin Sheehy, and Brian Sniffen. Principles of remote attestation. *International Journal of Information Security*, 10(2):63–81, 2011.
- [15] George Coker, Joshua Guttman, Peter Loscocco, Amy Herzog, Jonathan Millen, Brian O’Hanlon, John Ramsdell, Ariel Segall, Justin Sheehy, and Brian Sniffen. Principles of remote attestation. *International Journal of Information Security*, 10(2):63–81, 2011.
- [16] Simon Frost, Thomas Fossati, and Giridhar Mandyam. Arm’s confidential compute architecture reference attestation token. Internet-Draft draft-ffm-rats-cca-token-03, March 2026.
- [17] Google LLC. Google cloud confidential computing: Confidential VMs, confidential GKE nodes, and confidential space attestation. Google Cloud Documentation, 2024.
- [18] Steve Grubb. The Linux audit subsystem. Linux kernel documentation, 2017.
- [19] Vivek Haldar, Deepak Chandra, and Michael Franz. Semantic remote attestation: A virtual machine directed approach to trusted computing. In *Proceedings of the 3rd USENIX Virtual Machine Research and Technology Symposium*, pages 29–41, Berkeley, CA, USA, 2004. USENIX Association.
- [20] Naohiro Hayashibara, Xavier Defago, Rami Yared, and Takuya Katayama. The φ accrual failure detector. In *Proceedings of the 23rd IEEE International Symposium on Reliable Distributed Systems (SRDS)*, pages 66–78, Los Alamitos, CA, USA, 2004. IEEE Computer Society.

- [21] Intel Corporation. Intel Trust Domain Extensions (Intel TDX) module base architecture specification. Technical Report 348549-002US, Intel Corporation, Santa Clara, CA, USA, January 2023.
- [22] Andreas Kogler, Jonas Juffinger, Salman Qazi, Yoongu Kim, Moritz Lipp, Nicolas Boichat, Eric Shiu, Mattias Nissler, and Daniel Gruss. Half-Double: Hammering from the next row over. In *Proceedings of the 31st USENIX Security Symposium*, pages 3807–3824, Berkeley, CA, USA, 2022. USENIX Association.
- [23] Platon Kotzias, Juan Caballero, and Leyla Bilge. How did that get in my phone? unwanted app distribution on Android devices. In *2021 IEEE Symposium on Security and Privacy (S&P)*, pages 53–69, Los Alamitos, CA, USA, 2021. IEEE.
- [24] He Liu, Stefan Saroiu, Alec Wolman, and Himanshu Raj. Software abstractions for trusted sensors. In *Proceedings of the 10th International Conference on Mobile Systems, Applications, and Services (MobiSys)*, pages 365–378, New York, NY, USA, 2012. ACM.
- [25] Microsoft Corporation. Microsoft Pluton: Security processor architecture for Windows-class platforms. Microsoft Pluton documentation and Project Mu firmware sources, 2023.
- [26] Microsoft Corporation. Event tracing for Windows (ETW) architecture. Microsoft Windows Driver Documentation, 2024.
- [27] Microsoft Corporation. Microsoft azure attestation service: Reference documentation. Microsoft Azure Documentation, 2025.
- [28] Daniel Moghimi. Downfall: Exploiting speculative data gathering. In *Proceedings of the 32nd USENIX Security Symposium*, pages 7179–7193, Berkeley, CA, USA, 2023. USENIX Association.
- [29] Kit Murdock, David Oswald, Flavio D. Garcia, Jo Van Bulck, Daniel Gruss, and Frank Piessens. Plundervolt: Software-based fault injection attacks against Intel SGX. In *Proceedings of the 41st IEEE Symposium on Security and Privacy (S&P)*, Los Alamitos, CA, USA, 2020. IEEE.
- [30] NVIDIA Corporation. NVIDIA Hopper confidential computing on H100 and H200 GPUs. NVIDIA Confidential Computing Whitepaper and Hopper Architecture Documentation, 2024.
- [31] Anders Rundgren, Bret Jordan, and Samuel Erdtman. Json canonicalization scheme (jcs). RFC 8785, 2020.
- [32] Mark Russinovich, Manuel Costa, Cédric Fournet, David Chisnall, Antoine Delignat-Lavaud, Sylvan Clebsch, Kapil Vaswani, and Vikas Bhatia. Toward confidential cloud computing. *Communications of the ACM*, 64(6):54–61, 2021.
- [33] Ahmad-Reza Sadeghi and Christian Stübke. Property-based attestation for computing platforms: Caring about properties, not mechanisms. In *Proceedings of the 2004 Workshop on New Security Paradigms (NSPW)*, pages 67–77, New York, NY, USA, 2004. ACM.
- [34] Reiner Sailer, Xiaolan Zhang, Trent Jaeger, and Leendert van Doorn. Design and implementation of a TCG-based integrity measurement architecture. In *13th USENIX Security Symposium*, pages 223–238, Berkeley, CA, USA, 2004. USENIX Association.

- [35] Stefan Saroiu and Alec Wolman. I Am a Sensor, and I Approve This Message. In *Proceedings of the 11th Workshop on Mobile Computing Systems and Applications (HotMobile)*, pages 61–66, New York, NY, USA, 2010. ACM.
- [36] Secure Systems Lab. DSSE: Dead simple signing envelope (v1.0.2), 2024.
- [37] R. Sekar, Hanke Kimm, and Rohit Aich. eAudit: A fast, scalable and deployable audit data collection system. In *2024 IEEE Symposium on Security and Privacy (SP)*, pages 3571–3589, Los Alamitos, CA, USA, 2024. IEEE.
- [38] The Falco Authors. Falco: Cloud native runtime security. CNCF graduated project documentation, accessed 2026-05, 2025.
- [39] Jo Van Bulck, Marina Minkin, Ofir Weisse, Daniel Genkin, Baris Kasikci, Frank Piessens, Mark Silberstein, Thomas F. Wenisch, Yuval Yarom, and Raoul Strackx. Foreshadow: Extracting the keys to the Intel SGX kingdom with transient out-of-order execution. In *Proceedings of the 27th USENIX Security Symposium*, pages 991–1008, Berkeley, CA, USA, 2018. USENIX Association.
- [40] Y. C. Yeh. Triple-triple redundant 777 primary flight computer. In *Proceedings of the 1996 IEEE Aerospace Applications Conference*, volume 1, pages 293–307, Los Alamitos, CA, USA, 1996. IEEE.

A Open Science

The artifacts behind the formal and empirical claims are released for public inspection. This appendix names what is available, how to rebuild it, and how a reviewer recovers each load-bearing statement from its primary source.

A.1 Artifact availability

Three artifact classes accompany the paper. The Lean 4 mechanization that discharges the four theorems of Section 4 is the central artifact: a single paper-local source module of 589 lines, kept beside the paper rather than in the substrate’s proof root, and packaged as a self-contained Lake project together with the three substrate Treaty modules that form its import closure. The packaged copy is byte-identical to the module of record, and the supplementary package records the SHA-256 that a reviewer checks this against. The bibliography file lists primary-source citations to specifications, standards documents, and peer-reviewed papers. The paper text and the supplementary package are released under the same permissive license that governs the substrate’s public artifacts. For double-blind review the artifacts are submitted in anonymized form; a public repository location is named in the camera-ready version.

A.2 Reproducing the Lean mechanization

The Lake project pins a Lean toolchain through a checked-in `lean-toolchain` file, matching the release the substrate’s own proof root pins, and declares no external dependency. A reviewer checks the mechanization from a fresh extract:

```
tar xzf lean-source.tar.gz
cd chio-lean
```

```
lake build
lake env lean Chio/Treaty/SensorGroundedAdmission.lean
```

The `lakebuild` step builds the three substrate Treaty modules; the `lakeenvlean` step elaborates the paper-local module against them and exits with no output. The two steps stay separate because the module is checked exactly as it sits in the repository: from outside the root module, never registered in it. Against the substrate’s own proof root, the same check is `lakeenvlean` on `lean/SensorGroundedAdmission.lean` run from `formal/lean4/Chio`, and that is the run of record. Neither route reports a warning, and the sources contain no `sorry`. The `#print axioms` command run against each of the four theorems reports only the standard Lean kernel axioms `propext`, `Classical.choice`, and `Quot.sound`. No project-local axiom declaration appears anywhere in the dependency tree. The supplementary package’s `proof-manifest.toml` records the fully qualified theorem names, their source modules, and their axiom sets for tool consumption, and records that the module is paper-local, outside the substrate’s own theorem inventory.

A.3 Reproducing the empirical claims

The empirical claims of Section 6 rest on the sensor-state attestation field carried by every signed receipt produced by the admission kernel described in Section 5. A reviewer reproduces the figures by exercising the kernel against the receipt corpus described in Section 5, extracting each receipt’s signed attestation block, and verifying both the attestation signature and the DSSE subject digest that binds attestation bytes to body bytes. The canonical-JSON parser and the provider-record schema are part of the released runtime package. The runtime’s deployment identity is anonymized for review and named in the camera-ready version.

A.4 Bibliography accessibility

Every load-bearing citation resolves to a primary source that is publicly retrievable. The bibliography includes vendor specifications (Intel TDX, AMD SEV-SNP, Apple Private Cloud Compute, Arm CCA, AWS Nitro), Internet standards (RFC 8785, RFC 9334), and peer-reviewed publications from USENIX, IEEE, and ACM venues. No paywalled proceeding or unavailable technical report carries a load-bearing claim.

B Ethics Considerations

This appendix describes the ethical posture of the work in the categories USENIX Security policy names for the Ethics Considerations section.

B.1 Human and animal subjects

No human-subjects research and no animal-subjects research were conducted. No IRB review or animal-care protocol was required. The empirical evaluation in Section 6 is over receipts produced by a software admission kernel; the measured artifacts are signed canonical-JSON records and verifier-side mutation outcomes, not data derived from human participants. No personally identifying information was collected, analyzed, or stored at any point in the construction or evaluation.

B.2 Responsible disclosure

The sensor-grounded admission construction does not introduce new vulnerabilities. The construction is a verifier-side admission discipline that consumes existing trusted-execution-environment attestation primitives whose security properties are inherited from the underlying substrate. No new attacks against attestation primitives, signing keys, sensing hardware, or cryptographic constructions are proposed.

The TEE-compromise literature cited in Section 9 describes attacks that were disclosed by their original authors, assigned CVE identifiers, and (in most cases) mitigated by hardware vendors through microcode updates or successor generations: Plundervolt [29], Foreshadow [39], Downfall [28], and Half-Double [22] are each cited in their published, peer-reviewed forms. The paper acknowledges that TEE compromise reduces the construction to single-key signing for the duration of a compromise window, and names this reduction as a known structural boundary rather than as a new finding. No responsible-disclosure timeline was triggered by this work, and no embargo applies to any artifact in the submission.

B.3 Dual-use considerations

The construction’s primary beneficiary is the verifier, that is, the polity admitting receipts under a constitution. An attacker who controls the agent producing a receipt body gains nothing from the substrate attesting its own sensing posture: the attestation is adversarial information from the attacker’s perspective, since it constrains rather than enables admission. Forging an attestation requires compromising the TEE-rooted attestation signing key, which is the threat-model boundary named in Section 1 and discussed in Section 9.

The marginal dual-use concern is observability: a verifier sees more about the signing kernel’s substrate state than under the unstrengthened predicate, and an adversary positioned in the verifier role inherits that observability. This concern is intrinsic to any attested-system design that exposes substrate state to relying parties, and is not unique to the sensor-grounded construction. The construction does not introduce a novel attack surface, novel side-channel, or novel cryptographic primitive whose misuse would benefit an attacker.

B.4 Conflicts of interest

Following USENIX Security double-blind review policy, conflict-of-interest declarations are deferred to the camera-ready version. The authors will disclose any relationships requiring declaration under USENIX policy at that stage. The work does not benefit from any commercial relationship whose disclosure is foreclosed by the anonymous-review posture.